

Well Schematic 3D — Architecture & 3D Stack

Technical documentation

App Architecture Overview

Well Schematic 3D is an Angular 18 application that visualizes J-hook fracking well trajectories in 3D. It uses a frontend-only setup with an optional Express backend.

Main Components

WellSchematicPageComponent: Main page; holds params and wires the 3D viewer and sidebar

WellSchematic3dComponent: Three.js 3D viewer; builds and renders the wellbore

ParamsSidebarComponent: Sidebar for editing well parameters (depths, casing, perforations, etc.)

TrajectoryService: Computes the J-hook trajectory from parameters

WellSchematicParams: Model for all well parameters

3D Stack & Libraries

The 3D rendering is powered entirely by Three.js and its examples:

- THREE — Core library
- OrbitControls — `three/examples/jsm/controls/OrbitControls.js` for pan, zoom, rotate

Three.js Usage in the App:

1. Scene setup: **THREE.Scene**, **THREE.PerspectiveCamera** (with `logarithmicDepthBuffer`), **THREE.WebGLRenderer** (antialias, shadows, clipping)
2. Geometry: **TubeGeometry** (wellbore path), **CatmullRomCurve3** (smooth path), **CylinderGeometry**, **SphereGeometry**, **TorusGeometry**, **ConeGeometry**, **BoxGeometry**
3. Materials: **MeshStandardMaterial** with `clippingPlanes` for cutaway views
4. Lighting: **AmbientLight**, **DirectionalLight** (with shadows), **PointLight**
5. Cutaway: **THREE.Plane** for clipping, `renderer.localClippingEnabled = true`

Trajectory Math

TrajectoryService uses the minimum curvature method to compute the J-hook path: vertical segment !' build curve (arc) !' horizontal tangent. Output is `TrajectoryPoint[]` with `measuredDepth`, `trueVerticalDepth`, `inclination`, `azimuth`, `x`, `y`, `z`.

Data Flow

```
ParamsSidebarComponent (user edits)
  !' paramsChange.emit()
    !' WellSchematicPageComponent.onParamsChange()
      !' params = newParams
        !' WellSchematic3dComponent [params] input
          !' ngOnChanges !' buildWell()
```

Dependencies Summary

Angular (18.2.x): Framework, routing, forms

Three.js (0.183.2): 3D rendering

@types/three (0.183.1): TypeScript types
RxJS (7.8.x): Reactive utilities
zone.js (0.14.10): Angular change detection

There are no other 3D libraries — the 3D behavior comes entirely from Three.js and its built-in examples.